VP-40 variable-speed heat pump – Owner's quick guide

This guide covers daily operation of the VP-40 air-to-water heat pump as installed by Nordlicht Haustechnik. The full manufacturer manual is in the folder handed over at commissioning.

Display and modes

The controller has three modes: Auto, Heating only and Hot water only. In Auto the unit heats the house and the hot water tank as needed; this is the recommended setting. The display shows the current flow temperature, the outdoor temperature and the hot water tank temperature.

Recommended settings

Heating curve: 0.6 for radiators, 0.4 for underfloor heating. Hot water target temperature: 50 °C. A weekly anti-legionella run to 60 °C is set automatically on Sunday night. Raising the hot water target above 55 °C increases electricity use noticeably.

Error codes

E01 Low water pressure: check the pressure gauge, it should read between 1.0 and 2.0 bar. Top up using the filling valve under the hydraulic module. If the pressure drops again within days, call for service.

E05 Outdoor unit fan blocked: switch the unit off at the main switch, remove leaves or snow from the fan grille, switch on again. If the error returns, call for service.

E12 Communication error between indoor and outdoor unit: switch the unit off at the main switch for one minute and on again. If the error persists, call for service. Do not open the units.

E20 Sensor fault: the unit continues in a safe mode with reduced output. Service is needed within a few days.

Maintenance by the owner

Keep the outdoor unit free of leaves, snow and vegetation with at least 30 cm clearance on all sides. Clean the indoor air filter of the ventilation module every three months. Everything else is covered by the annual maintenance visit.

Winter operation

The outdoor unit defrosts itself; steam from the unit in cold weather is normal. Never switch the unit off for longer periods in winter, as the frost protection only works while it is powered. If you are away, use the Holiday mode instead, which keeps the house at 15 °C.

Nordlicht Haustechnik GmbH is a fictional company. This document exists to demonstrate a document-based support assistant.